MARCIN FIRMUGA



Python Developer | System Architecture

Self-taught software engineer with 680 hours development experience. 39,000+ lines of Python code. Specialization in system monitoring, real-time data processing, and API integration.

GitHub LinkedIn Website Email X (@hck_lab)

CAREER TRANSITION: IT TECHNICIAN TO SOFTWARE DEVELOPER

Self-taught Python developer transitioning from IT support role (2021-2022 internship) to full-time software development. Currently building PC_Workman, a system monitoring application in active development. Technical focus: real-time data collection, Windows system APIs, multithreaded architecture, and external API integration. Seeking junior developer position to contribute to production software projects.

680+	4	39k+	60
HOURS CODED	COMPLETE REBUILDS	LINES OF CODE	GITHUB CONTRIBUTIONS

CAREER JOURNEY

- **2018-2019**
  Game Translator & Localization | Black Deck Crew - Translated game dialogues, narratives, and technical documentation (ENG/PL)
- **2019-2023**
  Technical School (Technik Informatyk) | Zespół Szkół Ponadpodstawowych im. Stanisława Staszicy – IT technician education, hands-on system administration
  - **2020-2023 (DURING SCHOOL)**
    Lead Photographer & Photo Editor – One of two primary photographers for ceremonial and event photography. Managed post-processing, color correction, archival.
  - **2021-2022 (SCHOOL INTERNSHIP)**
    IT Support Intern | IT-M Kasy Fiskalne (Radom) – PC repair, system administration, technical support for fiscal registration systems
- **2024 (JANUARY)**
  Self-taught Python. Started learning system architecture and API integration.
- **2024 (SEPTEMBER)**
  v1.5.0 dashboard redesign. Began 4th major rebuild.
- **2024 (DECEMBER)**
  Integrated GPT-4 API for system diagnostics module (hck_GPT).
- **2026 (JANUARY)**
  v1.6.1 release. Dedicated website and GitHub documentation published.
- **2026 (Q2 TARGET)**
  v2.0 development - planned features include local ML models and advanced analytics.

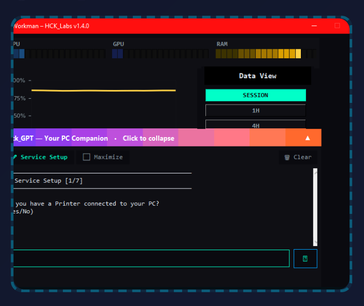
PERSONAL PROJECT: PC_WORKMAN

v1.6.1 (Alpha)

📊 LIVE DASHBOARD



HCK_GPT



System monitoring application for Windows providing real-time performance metrics, thermal data collection, and analysis capabilities. Implements multithreaded architecture for concurrent data collection. Integrates GPT-4 API for system diagnostics interpretation. Open source (MIT License).

IMPLEMENTED FEATURES

SYSTEM DIAGNOSTICS (HCK_GPT)

GPT-4 API integration providing human-readable explanations of system metrics and anomalies.

PERFORMANCE ANALYTICS

Correlation analysis of system metrics - CPU utilization, thermal data, GPU metrics, memory pressure.

ANOMALY DETECTION

Automatic identification of voltage spikes and thermal anomalies with historical logging.

HISTORICAL DATA ANALYSIS

Time-series navigation allowing review of historical system states and corresponding metric values.

TECHNOLOGY STACK

Python (Expert) · Tkinter GUI · Matplotlib · PyInstaller · Windows API · GPT-4 API · Multithreading · System Architecture





TECHNICAL SKILLS

PYTHON & ARCHITECTURE

- OOP & modular design (component registry)
- Multithreading & performance optimization
- Data logging (CSV, JSON profiles)
- Real-time system monitoring

EXTERNAL API INTEGRATION

- GPT-4 API integration and response handling
- REST API consumption and data parsing
- Windows WMI API (real-time system data)
- Error handling and API rate limiting

UI/UX & VISUALIZATION

- Tkinter GUI (4 major redesigns)
- Matplotlib data visualization
- 40% information density improvement
- Dark mode design systems

DEVOPS & DEPLOYMENT

- PyInstaller (.exe packaging)
- Git & GitHub (CI/CD ready)
- Version management (v1.0 - v1.6.1)
- GitHub Pages (Jekyll)

SYSTEM ENGINEERING

- Windows WMI API integration
- Real-time thermal monitoring
- Fan curve control & optimization
- Performance bottleneck detection

POSITION SOUGHT

Junior Python Developer | Software Engineer

Remote preferred. Full-time, internship, or contract roles considered.

Contact and Portfolio Links

GitHub LinkedIn PC_Workman Project Medium Articles Email

© 2025 Marcin Firmuga - Python Developer Portfolio